

Find the Maximum and Minimum Elements in a List

Write a Python function to find the maximum and minimum elements in a given list.

Input: [3, 1, 4, 1, 5, 9]

```
def find_max_min(numbers):
    maximum = numbers[0]
    minimum = numbers[0]

    for num in numbers:
        if num > maximum:
            maximum = num

        if num < minimum:
            minimum = num

    return maximum, minimum
```

```
numbers = [3, 1, 4, 1, 5, 9]
```

```
maximum, minimum = find_max_min(numbers)
```

```
print("Maximum:", maximum)
print("Minimum:", minimum)
```

Output:

Maximum: 9

Minimum: 1

Remove Duplicates from a List

Write a Python function to remove duplicates from a list while preserving the order.

Input: [1, 2, 2, 3, 4, 4, 5]

```
def remove_duplicates(numbers):
    result = []

    for num in numbers:
        if num not in result:
            result.append(num)

    return result
```

```
numbers = [1, 2, 2, 3, 4, 4, 5]
```

```
print(remove_duplicates(numbers))
```

Output:

```
[1, 2, 3, 4, 5]
```

Find the Intersection of Two Lists

Write a Python function to find the intersection of two lists.

Input: [1, 2, 3, 4], [3, 4, 5, 6]

```
def find_intersection(list1, list2):
```

```
    result = []
```

```
    for num in list1:
```

```
        if num in list2 and num not in result:
```

```
            result.append(num)
```

```
    return result
```

```
list1 = [1, 2, 3, 4]
```

```
list2 = [3, 4, 5, 6]
```

```
print(find_intersection(list1, list2))
```

Output: [3, 4]